

Project Closeout Report

Add More and Easier Levels to Cardano Capture the Flag (CTF)

Key project details:

- **Project name:** Add more and easier levels to Cardano Capture the Flag (CTF) ([Project link](#))
- **Project number:** 1200018
- **Project manager:** Michal Porubsky
- **Project start:** August 2024
- **Project completion:** January 2026
- **Closeout video:** <https://youtu.be/Co88mS-qOal>

Project summary

We have successfully delivered 14 new levels to the Cardano Capture the Flag (CTF) educational game, creating a complete “Banking Series” that provides a gentler, more progressive introduction to Cardano smart contract security. This new series builds security intuition gradually through a simple bank system that evolves level by level, addressing feedback from the original CTF that the learning curve was too steep.

The project delivered all proposed milestones: 14 new smart contract vulnerability challenges, comprehensive documentation, four Medium blog posts covering the new content, and a final hints and solutions blog post. All implementations are tested on Cardano Preview Testnet and the codebase was modernized by migrating from Deno/Lucid to Node.js/Lucid Evolution for better maintainability as a bonus.

Challenge KPIs

The Cardano Open: Developers challenge set several KPIs relevant to our project:

Increase the number and quality of open-source projects. We have added 14 new open-source smart contract challenges to the Cardano CTF repository, bringing the

total to 25 educational levels (11 original + 14 bank series). Each level includes thoroughly documented on-chain validators written in Aiken, off-chain interaction code using Lucid Evolution, and comprehensive README files explaining the tasks and hinting at the vulnerability in case of the banking series. All materials are published under open-source GPL-3.0 license on GitHub. The quality of the project is evidenced by it being mentioned in the official Cardano Developer Documentation, under the [Security](#) section.

Increase adoption of Cardano technology. The Banking Series specifically addresses a barrier to adoption identified in our previous CTF: **complexity**. By providing progressively harder challenges starting from fundamental programming concepts and gradually introducing Cardano-specific patterns, we lower the barrier for developers new to the ecosystem. The series serves as a bridge between general programming knowledge and the advanced security concepts required for safe Cardano development.

Open source license requirement. All project materials are published under GPL-3.0 open source license on GitHub, freely available for anyone to use, learn from, and contribute to.

High quality documentation delivery. Each of the 14 new levels includes:

- A detailed README explaining the challenge context and objectives.
- Commented smart contract code in Aiken.
- Template player files with base64-encoded hints, ready to be modified and extended to exploit the vulnerability.
- Complete off-chain code demonstrating proper contract interaction.
- Tests verifying successful vulnerability exploitation.

Project KPIs

Our project successfully delivered all proposed milestones and objectives:

Milestone Deliverables:

- **✓ 14 New CTF Levels:** The complete Banking Series (bank_00 through bank_13) covering:
 - Levels 0–1: Core invariants and fundamental security checks.
 - Levels 2–5: UTxO-specific patterns (mutual exclusion, lifecycle, configuration).

- Levels 6–9: Token security (minting policies, double satisfaction, self-collision).
- Levels 10–13: Advanced patterns (refactoring pitfalls, signature schemes, replay attacks).
- ✓ **Blog Posts:** Four Medium articles published:
 - [Banking on Cardano: A CTF Journey Begins](#) — Introduction to the series philosophy.
 - [When Simple Gets Interesting: Tokens Enter the Banking Series](#) — Covering intermediate level patterns.
 - [Deja Vu and Cheque Chaos: Banking Levels 8–13](#) — Covering cheque-based vulnerabilities.
 - [Banking Series CTF — Hints and Solutions](#) — Comprehensive hints and solutions guide.
- ✓ **Framework Modernization:** Migrated from Deno/Lucid to Node.js/Lucid Evolution for improved maintainability and developer experience.
- ✓ **Testnet Validation:** All levels tested on Cardano Preview Testnet with on-chain solution recording.

Engagement and Adoption Indicators:

- **GitHub Repository:** The [cardano-ctf](#) repository continues to receive engagement with stars (currently at 34), forks (12), views (100 in last 14 days alone!), and git clones (37 in the last 14 days!). We consider this a major success.
- **Discord Community:** Discord channel for players to ask questions, share solutions, and discuss security topics. There's not been much activity there, but the announcements are there, new people keep coming and there's been a few messages here and there.
- **On-chain Scoreboard:** Successful solutions are recorded on the Preview Testnet, with a scoreboard script displaying overall number of solutions per level. So far, the banking series was properly (on testnet as well) solved by 1 non-team person only — always immediately upon new tasks being announced. That's not a lot, but the tasks have just recently been added and we believe it will grow progressively as now the series is finalized and more people are incentivized to go into it — as we have seen with the previous series, it keeps growing. We'll try to do more marketing on it and get people to try it.
- **Social Media:** Regular updates on X and promotion of new levels and blog posts.

- **Cardano Developer Mention:** The CTF is mentioned in the Cardano Developer Documentation, under the [Security](#) section highlighting its unique role in learning about security of Cardano smart contracts.

Technical Achievements:

- Created a progressive learning path from basic programming errors to advanced Cardano-specific vulnerabilities.
- Demonstrated how simple banking functionality can harbor 14 distinct vulnerabilities.
- Showed the iterative nature of security: each fix potentially introduces new attack vectors and every refactor can be lethal.
- Modernized the entire codebase for better long-term maintainability.

Key Achievements

Progressive Learning Design: The Banking Series addresses feedback from our original CTF that the learning curve was too steep. By starting with vulnerabilities that would be bugs in any programming language and gradually introducing Cardano-specific concepts, we created an accessible on-ramp for developers at all experience levels.

Interconnected Narrative: Unlike standalone challenges, the Banking Series tells a story. Each level's vulnerability arises from attempting to fix the previous one, mirroring real-world development cycles. Players experience firsthand how security patches can introduce new attack vectors, building intuition that no amount of reading can provide.

Comprehensive Vulnerability Coverage: The 14 levels cover a broad spectrum of vulnerabilities:

- Missing bounds checks and core invariants.
- Datum manipulation and ownership hijacking.
- Mutual exclusion failures between code paths.
- UTxO lifecycle misunderstandings.
- Configuration and reference input attacks.
- Minting policy weaknesses.
- Interesting double satisfaction and self-collision patterns.

- Signature vulnerabilities and replay attacks.

“Fix-It” Challenge Philosophy: We strongly encourage players to fix each vulnerability after exploiting it, then compare their solution with the next level’s implementation. This approach pushes the challenge to the next level, as it forces players to think of remediation strategies for the vulnerabilities found in the previous levels as well.

Key Learnings

Team Dynamics and Timely Delivery: The project faced significant delivery challenges due to team reshuffling. One team member left during the project, and another took a 6-month sabbatical, which required us to submit a change request and ultimately delayed completion. This kind of black-swan event reinforced the importance of building resilience into project timelines when working with small teams, as personnel changes can have outsized impacts on delivery schedules. Despite these setbacks, we maintained our commitment to quality over rushing to meet original deadlines.

Organic Reach of Quality Educational Content: Despite suboptimal marketing efforts, the CTF has generated consistent organic traffic. People continue to discover and use it as learning material on an ongoing basis. We’ve heard from developers at conferences who mentioned using it for their security education, and notably, one unrelated person solved levels immediately upon release—demonstrating that the content resonates with its target audience. While marketing could certainly be improved, this experience validates that quality educational material eventually finds its audience and gains broader attention through word-of-mouth and community sharing. We expect this trend to continue as the complete Banking Series becomes more widely known — especially since now it’s part of the Cardano Developer docs as well.

Next Steps

Repository Maintenance: We will continue to maintain the Cardano CTF GitHub repository, addressing any issues, and incorporating community feedback.

Community Engagement: Our Discord channel remains open for players to ask questions, share solutions, and discuss security topics. We continue to engage with the community on X and other platforms.

Potential Extensions: The Banking Series framework could be extended with:

- Additional vulnerability patterns as new attack vectors are discovered.
- Multi-validator scenarios bridging to the original CTF's complexity.
- Possibly including advanced concepts as vulnerabilities coming from CIP-113 programmable assets, or other upcoming standards and features as they are released.
- Possibly add levels in different contract languages.
- Similar framework-ed development-first projects, guiding developers to build (as opposed to exploit) vulnerability-proof smart contracts.

Final Thoughts

The Banking Series CTF represents our commitment to improving Cardano ecosystem security through education. Every catastrophic hack in blockchain history started with someone missing a “simple” check. By making security education accessible and engaging, we hope to prevent vulnerabilities before they make it to production.

The project delivered all proposed milestones: 14 new levels, multiple blog posts, framework modernization, and comprehensive documentation. More importantly, we created a learning tool that will continue to serve the community for years to come.

We thank Project Catalyst and the Cardano community for their support. Security is a shared responsibility, and educational resources like this CTF are essential infrastructure for a healthy ecosystem.

Links

- **GitHub Repository:** <https://github.com/Invariant-0/cardano-ctf>
- **Medium Publication:** <https://medium.com/@invariant0>
- **Discord:** <https://discord.gg/5XVW2MUdWu>
- **X:** https://x.com/invariant0_com
- **Invariant0 Website:** <https://invariant0.com/>
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